

It's a Match!

Feature Optimization for Entity Matching in Economic History

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Abstract

This paper demonstrates that feature optimization, the joint expansion and targeted reduction of similarity measures, improves entity matching in economic history. Across three use cases, we show that simultaneously employing a wide range of (up to 24) similarity concepts improves performance across all tested classifiers (Logit, Probit, Gradient Boosting, ANNs), and that for linear classifiers a subsequent feature selection step further increases performance in high-dimensional settings. We accompany the methodology with an open-source package providing a unified interface for training, storing, and reusing models. Matching performance, our results show, depends primarily on similarity-concept choice rather than on model complexity.

1 Introduction

Entity matching is a fundamental computational technique for leveraging the growing availability of increasingly granular data across the social sciences. The field of economic history has been particularly active in developing and enhancing such techniques, establishing entity matching as a distinct strand of methodological research (e.g., [Feigenbaum, 2016](#), [Eriksson, 2015](#), [Abramitzky et al., 2020](#), [Helgertz et al., 2022](#), [Goeken et al., 2011](#), [Buckles et al., 2025](#)). These methodological advances, in turn, provide the technical foundation for scholars to address questions that were previously out of reach given the resource intensity of manual record matching ([Gutmann et al., 2018](#)).

Despite this progress, there remains limited guidance on how researchers should select, configure, and optimize entity-matching procedures across different empirical settings. This paper addresses this gap by introducing a flexible, domain-agnostic methodology that adapts feature construction to the idiosyncrasies of historical data. Our proposed methodology is based on feature optimization, the process of extracting the strongest possible signal from the underlying data to maximize the information available to the classifier.

Our methodology implements this idea in two stages: first, by inflating the number of similarity measures applied to each field pair (*feature expansion*), and then by removing

highly correlated and uninformative features (*feature selection*).¹ This two-stage procedure identifies the optimal bundle of features for a given matching task without facing problems of multicollinearity.

Our methodology eliminates the need for researchers to pre-select and manually tune a small number of similarity measures, an approach common in applied work. Instead, the methodology allows a supervised classifier to learn how similarity measures should be weighted and combined, transforming what is often a heuristic choice into a reproducible and flexible procedure. Importantly, the methodology builds on matching techniques established in economic history rather than relying on word embeddings commonly used in computer science. This makes it robust to the noisy and archaic language typical of historical sources, which frequently causes out-of-vocabulary problems when applying embedding-based approaches.

The importance of these issues is evident in the widespread use of matched datasets in recent empirical work. Scholars have moved from running ordinary least squares regressions on a 5 percent sample of the U.S. Census ([Anderson, 1997](#)) to performing a wide range of computational analyses on full Census samples in recent decades. Similarly, the increasing scale and quality of matched census samples have enabled more sophisticated analysis of topics such as intergenerational social mobility and long-run inequality (e.g., [Long, 2013](#), [Long and Ferrie, 2013](#), [Abramitzky, Boustan, Jacome and Perez, 2021](#), [Berger et al., 2023](#), [Olivetti and Paserman, 2015](#), [Clark and Cummins, 2015](#)), the returns to migration (e.g., [Abramitzky et al., 2014](#), [Dribe et al., 2022](#), [Andersson, 2025](#)), and the effects of technological change on labor market outcomes (e.g., [Feigenbaum and Gross, 2024](#), [Gray, 2013](#)). Moreover, matching of patents has made it possible to trace the determinants of innovation over time (e.g., [Kogan et al., 2017](#), [Moser et al., 2014](#), [Bell et al., 2019](#)), while matched firm-level datasets have enabled the study of financial shocks, organizational structure, and firm performance (e.g., [Frydman et al., 2015](#), [Berger and Ostermeyer, 2025](#)).

¹Importantly, we define a "feature" as the result of applying a single similarity measure to a field pair. Consequently, multiple features can be derived from the same field pair when more than one similarity measure is applied. For an intuitive illustration of the two central components of feature optimization, see Figure I.

At the same time, the growing reliance on algorithmic entity matching in the data-generating process in economic history has raised concerns about the reliability and comparability of the resulting datasets. In particular, [Ruggles et al. \(2018\)](#) emphasizes the trade-off between errors of omission (missed matches) and errors of commission (false matches), noting that false matches can systematically bias estimates of social- and occupational mobility, by causing attenuation bias. Although, conservative, probabilistic, and machine-learning-based approaches can substantially reduce false positives, they do so by accepting lower match rates ([Abramitzky, Boustan, Eriksson, Feigenbaum and Pérez, 2021](#)). In addition to selecting the optimal classifier for a given use case, researchers must also make explicit decisions about the matching procedure. Among those are the choice of appropriate similarity measures (e.g., those based on the Jaro–Winkler or Levenshtein distances), acceptable similarity thresholds, which fields to compare, and what training data to use ([Feigenbaum et al., 2025](#)). Consequently, the choices researchers make when constructing matches have direct implications for how scholars reconstruct life courses, occupational trajectories, firm dynamics, and patterns of social mobility across time. Errors introduced at the matching stage can potentially propagate into stylized facts about industrialization, inequality, and structural change.

Against this backdrop, we evaluate the utility of the proposed methodology across three distinct matching applications, using heterogeneous data sources to demonstrate its flexibility. We show that feature optimization improves matching performance in two ways: through systematic feature expansion and through targeted feature reduction. First, expanding the set of field-level similarity measures consistently enhances performance across both linear and non-linear classifiers. Second, removing redundant and weakly informative similarity measures further improves results for linear models. Together, these findings demonstrate that performance in entity matching is improved by incorporating a rich set of similarity measures. One important result is that primarily linear classifiers experience improvement in performance from feature selection, while more complex models such as Artificial Neural Networks (ANN) do not. This result underscores that increased model complexity does not necessarily translate into improved matching

performance. To facilitate adoption and replication, the proposed methodology is implemented as an open-source Python package, **neer-match**, providing researchers with a unified and standardized interface for training, storing, and reusing matching models across heterogeneous historical data sources.

This paper is structured as follows. Section 2 situates our methodology within the computer science and economic history literatures on entity matching, highlighting the gap between performance-oriented advances and the inferential requirements of historical research. Section 3 introduces the concept of feature optimization and formalizes our two-stage expansion and selection procedure. Section 4 describes the implementation of the methodology and evaluates it across three use cases. In addition, it discusses the heterogeneity of matching performance across use cases with different matching classifiers. Section 5 concludes by discussing the broader implications for methodological practice in economic history and related fields.

2 Previous Literature

Entity matching has developed along several methodological lines depending on the properties of the data to be matched and the research objectives. One line of work focuses on constructing longitudinal datasets from noisy and incomplete sources, which is common when working with historical data. In this tradition, the choice of which similarity measures to apply to which field pairs, as well as other modeling decisions such as classifier type or matching thresholds, is often guided by convention or heuristic reasoning. There is limited systematic evaluation of how alternative specifications affect performance in a given empirical setting. Alongside this tradition, more automated matching frameworks have been developed for large-scale and standardized data environments. These approaches typically emphasize predictive performance and rely on techniques such as word embeddings, which can be less suitable for heterogeneous and historically specific data. Over time, insights from these frameworks have increasingly been absorbed into historical matching, even as some aspects of the matching process, notably the selection of features, have remained largely informal.

2.1 *Matching Historical Sources: Methods and Evolution*

Within economic history, entity matching has been adopted across a range of substantive domains. Among these domains, the census-matching literature has emerged as particularly methodologically influential, shaping how matching is performed and evaluated. As such, the quality of matching is evaluated not primarily in terms of benchmark accuracy, but in terms of its implications for downstream historical inference. False matches are treated as a source of measurement error that can potentially bias estimates, while missed matches are a source of concern for the representativeness of the matched sample. Methodological development has therefore been closely tied to issues related to sample selection and the stability of substantive results under the utilization of alternative matching strategies (e.g., [Bailey et al., 2020](#), [Ghosh et al., 2024](#)).

In other words, models that cannot be easily audited and retrained on alternative samples face substantial barriers to adoption, even when they may outperform less demanding methods on technical benchmarks. This is not to say that advances from computer science have not been absorbed into historical scholarship. Indeed, several algorithmic improvements have been widely adopted. Among the most widely adopted approaches are blocking strategies ([Michelson and Knoblock, 2006](#), [Christen, 2012](#)), probabilistic field weighting in the tradition of [Fellegi and Sunter \(1969\)](#), and supervised classification methods for matching decision rules ([Feigenbaum, 2016](#), [Goeken et al., 2011](#)).

In the same vein, the methodological evolution of entity matching for historical sources has progressed through a series of increasingly formalized yet still largely interpretable modeling strategies. Early contributions relied on hard-coded rules for probabilistic matching ([Ferrie, 1996, 2005](#)), later extended by expectation–maximization frameworks ([Abramitzky et al., 2020](#)), logit and probit models ([Feigenbaum, 2016](#), [Helgertz et al., 2022](#)), and, more recently, gradient-boosting approaches ([Price et al., 2021](#)). These approaches have been systematically evaluated, with results showing heterogeneous levels of accuracy across populations, time periods, and source materials ([Bailey et al., 2020](#), [Abramitzky et al., 2020](#), [Ghosh et al., 2024](#)). However, across these approaches, feature selection has typically remained somewhat ad hoc, often guided by convention or

incremental adjustments based on inductive reasoning.

Beyond census matching, similar patterns characterize entity matching in other domains of economic and financial history whenever stable unique identifiers are absent. Researchers have frequently combined deterministic matching rules with substantial manual verification (e.g., [Florackis et al., 2023](#), [Bartram et al., 2022](#), [Persson, 2020](#), [Frydman et al., 2015](#), [Bell et al., 2019](#)), while rule-based approaches relying on string similarity thresholds have been and also remain common (e.g., [Moser et al., 2014](#), [Cunningham, 2018](#), [Berger and Ostermeyer, 2025](#)). At the same time, the utilization of supervised learning algorithms in the census-matching literature have increasingly been adapted to other strands of historical research that require matching between sources (e.g., [Buggle et al., 2023](#), [Rijpma et al., 2020](#)). In practice, however, these varying methodological approaches are not sharply distinct. Many applications employ multi-stage matching procedures that combine deterministic blocking, string similarity metrics such as Jaro–Winkler distance, and, increasingly, supervised classifiers that weight these similarity features to reduce false positives and improve matching quality.

2.2 What Makes Historical Data Hard for Algorithmic Matching

Entity matching has been studied for decades (e.g., [Dunn, 1946](#)). Early approaches treated relevance as an intrinsic property of individual variables, ranking features according to criteria such as mutual information or correlation with the target outcome ([Peng et al., 2005](#)). Regularization-based methods, most prominently the LASSO, formalized this logic within a predictive framework by shrinking weakly informative coefficients toward zero and thereby selecting a parsimonious subset of features ([Tibshirani, 1996](#), [Zhao and Yu, 2006](#)). More recent developments have shifted attention toward model-based notions of relevance. In tree ensembles, such as random forests and gradient boosting, variable importance is inferred from a feature’s contribution to recursive partitioning and predictive gains across an ensemble of models ([Breiman, 2001](#), [Chen, 2016](#)). Neural network-based approaches extend this logic further, as relevance is expressed through learned latent representations rather than through explicit, field-level

effects, making feature contributions inherently dependent on the model architecture and training dynamics (Bengio et al., 2013, LeCun et al., 2015). Especially over the past twenty years, the field has produced a rich set of approaches ranging from rule-based and crowdsourcing methods to machine learning (ML), with deep learning (DL) now setting the performance benchmark.² Taken together, these developments reflect a shift from variable-based to model-based notions of relevance, where informative signals are learned from data rather than specified ex ante, often through embedding representations, active learning, or transfer learning techniques.

These ideas have begun to influence large-scale historical entity matching projects. For example, recent applications of gradient boosting to U.S. census data assess variable importance empirically in terms of its contribution to the accuracy of matching at scale, enabling the construction of hundreds of millions of individual matches across the period 1850–1940 (Buckles et al., 2025). This work illustrates how digital methods have expanded the range of historical questions that can be addressed at population scale. However, such applications typically rely on relatively well-defined and consistently recorded variables within structured administrative data. As a result, their broader transferability remains constrained by differences in the availability, definition, and reliability of fields across countries and time periods. In other words, despite substantial technical progress, these approaches are not necessarily straightforwardly applicable to the problems historians and social scientists face.

This lack of transferability arises because working with digitized historical sources entails challenges that these methods of matching were not designed to address. An increasing share of data in quantitative economic history is now extracted from historical sources using optical character recognition (OCR). OCR, however, is far from perfect and often introduces systematic transcription errors (Kettunen and Pääkkönen, 2016). As a result, in such settings, models that rely on word embeddings may perform poorly, since misspelled words are unlikely to be well represented in the embedding space (Ehrmann

²See, for instance, Low et al. (2024), Li et al. (2020), Mudgal et al. (2018), Wang et al. (2020), Qian et al. (2017), Konda et al. (2016), Bilenko and Mooney (2003), Ding et al. (2024), Li et al. (2021), Wu et al. (2020).

et al., 2023). A similar problem arises from the vocabulary itself: many terms, names, and corporate labels in historical sources are archaic or idiosyncratic and therefore absent from modern pre-trained embedding models, leading to severe out-of-vocabulary issues (Piotrowski, 2012). This poses particular challenges for modern entity matching algorithms, which have largely been developed and trained on contemporary datasets that are rich in variables and highly standardized. As a result, there is substantial variation in which fields provide sufficient discriminative signal for a given model, since systematic transcription errors and uneven field coverage can introduce noise rather than informative variation.

Beyond these technical issues, matching procedures do not operate on neutral data: matching errors and unmatched records are rarely distributed at random, and they disproportionately concentrate on historically marginalized groups, shaping the populations that become visible in historical analysis (Bailey et al., 2020, Goeken et al., 2011, Abramitzky et al., 2020). The linkage of women is a stark case in point, since surname changes at marriage break the standard matching keys; Cummins (2025) circumvents this by exploiting the completeness of English vital registration and restricting attention to records where exact name matching is unambiguous. Such institutional fixes, however, generalize poorly across the heterogeneous source materials that economic historians typically encounter.

Yet what unites historical matching applications is that similarity is not a stable property of records but depends critically on which fields are compared and how they are weighted within a given context. Historical records are frequently noisy and incomplete, and the structure of the records is shaped by evolving administrative practices and semantic drift. As a result, both the availability and the comparability of fields may vary across time, space and the structure of OCR-errors. The practical lever available to the researcher, in turn, lies not in the data itself but in how the available fields are constructed and combined.

Hence, one of the most pressing issues for the historian is to determine which dimensions of the historical record can be transformed into stable and discriminative signals

for matching. The central challenge, therefore, lies in selecting field pairs and similarity measures so as to maximize signal under conditions of transcription error, missingness, and semantic drift, and in selecting those features that contribute information without introducing spurious noise. In computer science, this problem is addressed under the heading of feature selection: the systematic evaluation of which input attributes contribute useful information to a predictive task and which instead introduce redundancy or noise.

Moreover, as our results indicate, performance does not necessarily improve with increasing model complexity. Rather, the primary determinant of matching quality is whether the most salient and informative features are included at all. This argument does not contradict the evidence, cited above, that ANNs, particularly DL architectures, often outperform simpler approaches on standard entity matching benchmarks or in settings where the idiosyncrasies of the data are well characterized. Rather, the point is that such models involve optimization over a substantially higher-dimensional parameter space, rendering their performance gains conditional on whether the model’s implicit assumptions about structure and regularity—that is, its inductive bias—align with the statistical properties of the data. As [Belkin et al. \(2019, p. 15850\)](#) emphasizes, model capacity alone does not determine generalization; what matters is how well the learned predictor matches the “inductive bias appropriate for the problem at hand”. In contexts where this alignment is weak (as is often the case with sparse or noisy data) greater model complexity does not necessarily translate into superior matching performance.

Taken together, these considerations suggest that progress in historical entity matching is less likely to come from further gains in algorithmic sophistication alone, and more from the development of methodologies that make the construction, evaluation, and selection of features explicit, transparent, and reproducible. While continued integration of more powerful algorithms holds clear promise, such advances must be accompanied by systematic approaches for diagnosing and mitigating the biases introduced by the particularities of historical source materials.

2.3 Formalizing Feature Optimization for Historical Matching

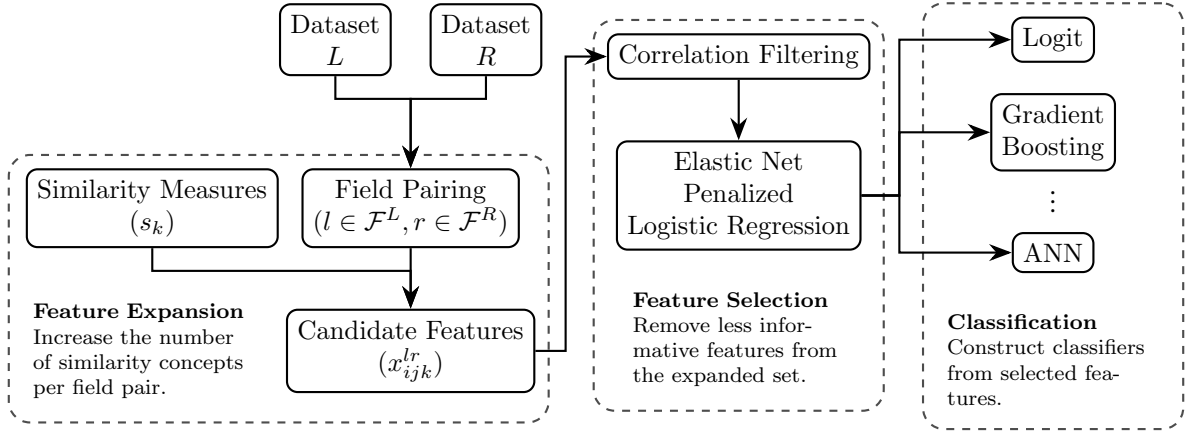
From this perspective, the contribution of this paper is not to introduce a new matching algorithm per se, but to provide a methodology that formalizes and systematizes a practice that is already implicit in historical work: the careful construction, testing, and exclusion of fields based on the consequences their inclusion has on the quality of matching for a given dataset.

Consequently, we draw on conceptual advances in feature optimization from computer science to develop a methodology tailored to address key challenges in entity matching for historical data. By formalizing feature optimization as a distinct, reproducible methodological step, the methodology makes explicit which features of the historical data provide independent discriminative information for the matching process and under what conditions their contribution remains stable across sources, and time periods.

3 Methodology

Historical entity matching has often treated the choices of similarity measures as a central methodological decision ([Abramitzky, Boustan, Eriksson, Feigenbaum and Pérez, 2021](#), [Ruggles et al., 2018](#)). Measures such as Jaro–Winkler, Levenshtein distance, and related token- or character-based measures are commonly adopted as default tools for comparing names and textual fields ([Adam et al., 2021](#)). However, as the intrinsic properties of data generation and extraction are distinct across historical data sources, the matching performance of similarity measure choices is context dependent. Thus, predefined collections of similarity measures are unlikely to constitute best choices for matching historical entities in different domain areas. Our feature optimization methodology systematically identifies combinations of fields pairs and similarity measures with optimal performance for matching entities from two given historical data sources. [Figure I](#) visualizes the feature optimization workflow. We document the details of the workflow in the following sections.

Figure I: Feature Optimization Workflow.



3.1 Feature Expansion

We consider entity matching scenarios involving two distinct structured datasets L and R (the two datasets may come from different volumes of the same source or from entirely different sources). Each dataset has a list of fields that we denote $\mathcal{F}^L$ and $\mathcal{F}^R$. The sets $\mathcal{F}^L$ and $\mathcal{F}^R$ have overlapping information that (i) identifies unique entities in the two datasets and (ii) is non-harmonized and potentially incomplete. For example, in the company matching application of Section 4, the two datasets are OCR extracted data from historical archives of listed companies with, non-harmonized across archives, fields such as company names, foundation dates, and headquarter addresses. The matching process combines fields from $\mathcal{F}^L$ and $\mathcal{F}^R$ to construct field pairs and assess whether two records, one drawn from L and one from R , refer to the same entity (left dashed box of Figure I).

A field pair lr combines a field $l \in \mathcal{F}^L$ and a field $r \in \mathcal{F}^R$. This combination gives up to $n = n^L \cdot n^R$ field pair observations, where n^L and n^R are the number of records in L and R , respectively. We denote as $\mathcal{L}^{(l)}$ (and $\mathcal{R}^{(r)}$) the field space of l (and r), namely the set of potential values for field l (r) in dataset L (R). Multiple similarity measures with distinct properties are compatible with a field pair.

Let m^{lr} be the number of similarity measures compatible with a field pair lr . For any $k = 1, \dots, m^{lr}$, the similarity measure $\mu_k: \mathcal{L}^{(l)} \times \mathcal{R}^{(r)} \rightarrow \mathbb{R}$ maps field pair values $(z_a^{(l)}, z_b^{(r)})$, where $a = 1, \dots, n^L$ and $b = 1, \dots, n^R$, to ML-compatible numerical scores

$s_{abk}^{lr} = \mu_k(z_a^{(l)}, z_b^{(r)})$. The scores vectors $\{s_{.k}^{lr}\}_{rlk}$ constitute the candidate features of the matching methods that our analysis examines. Our methodology provides a semi-automated selection of features (namely, choices of l , r , and k) that optimize the entity matching performance for the n records in the cross product of L and R .

3.2 Feature Selection

Feature expansion generates an inclusive collection of features to be used in entity matching. A naive expansion approach would involve all candidate features $\{s_{.k}^{lr}\}_{rlk}$ constructed in the extraction phase. Yet, this approach is prone to multicollinearity, which can be tolling for numerical stability in high-dimensional spaces such as those involved in entity matching, and overfitting. Hence, we employ a filtered expansion of the feature space to retain the most informative yet linearly independent features (middle dashed box of Figure I).

Let p the number of extracted features and let $\mathbf{S} \in \mathbb{R}^{n \times p}$ denote the matrix of extracted feature candidate scores. We compute the absolute correlation matrix $R_{\text{corr}} = [|r_{ij}|]_{i,j=1}^p$ among all p candidate features, where $|r_{ij}|$ is the correlation between the i and j features. In the first stage of the feature selection process, the features are then grouped into clusters of collinear variables. Two features i and j are grouped in the same cluster when $|r_{ij}|$ exceeds a (hyper-parameter) clustering threshold $\tau \in (0, 1)$. This generates q clusters that we denote as $\{\mathcal{C}_t\}_{t=1}^q$. Subsequently, we select a single candidate feature from each cluster.

Specifically, the feature in $\mathcal{C}_t$ with the highest absolute correlation with the binary match indicator $\mathbf{y} \in \{0, 1\}^n$ is retained, i.e., $j_t = \arg \max_{j \in \mathcal{C}_t} |r_{j,y}|$, and the remaining features in the cluster are discarded. This pre-selection reduces multicollinearity issues that can numerically destabilize the subsequent feature selection step. The pre-selection approach generates q features $\{\mathbf{x}_{(j)}\}_{j=1}^q$, with $\mathbf{x}_{(j)} \in \mathbb{R}^n$ and linearly independent observations.

In the feature selection's second stage, we estimate an elastic net-regularized logistic model, where the coefficients of weak or overlapping variables are shrunk toward zero

through penalization. The pre-selected features are standardized to zero mean and unit variance to avoid scaling issues. The selection model minimizes

$$Q(\boldsymbol{\beta}; \mathbf{X}, \mathbf{y}) = -\frac{1}{n} \sum_{h=1}^n w_{y_h} [y_h \log \hat{p}_h + (1 - y_h) \log(1 - \hat{p}_h)] + \frac{\alpha}{2} (1 - \lambda) \|\boldsymbol{\beta}\|_2^2 + \alpha \lambda \|\boldsymbol{\beta}\|_1,$$

where $\boldsymbol{\beta} \in \mathbb{R}^q$ is a vector of regression coefficients, $\hat{p}_h = \sigma(\mathbf{x}_h^\top \boldsymbol{\beta})$, $\alpha = 1/C$ is the penalty parameter, λ governs the balance between the ℓ_1 (lasso) and ℓ_2 (ridge) penalty components, and w_{y_h} are inverse-frequency class weights that address the severe class imbalance inherent in entity matching.

The hyper-parameters C and λ are calibrated jointly via stratified k -fold cross-validation over a grid of logarithmically spaced C values and a grid of linearly spaced λ values to optimize the average precision score. Finally, the model using best calibrated C and λ is fit on the full training set and features whose estimated coefficients are close to zero are excluded from the entity matching process. The class weights are computed as $w_c = \frac{n}{K \cdot n_c}$, where n is the total number of training pairs, $K = 2$ is the number of classes, and n_c is the number of pairs in class $c \in \{0, 1\}$. This up-weights the rare match class and down-weights the abundant non-match class, ensuring that the loss contribution of each class is proportional to its inverse frequency rather than its count.

3.3 Classification

Feature optimization, consisting of feature expansion and subsequent selection, identifies a sparse subset of $\{\mathbf{x}_{(j)}\}$ with features that contribute independent predictive information to the classification between matches and non-matches. Moreover, the weights $\{w_c\}_{c \in \{0,1\}}$ account for extreme class imbalances typical in entity matching, where true matches constitute only a small fraction of possible record pairs.

The entity matching classifiers are trained using the optimized set of features. The selected features can be combined with one or more binary classifiers (right dashed box of Figure I). The three use cases of Section 4 illustrate this by using and evaluating the entity matching performance of Logit, Probit, Gradient Boosting, and ANN classifiers.

4 Applications in Historical Companies, Persons, and Publications

We demonstrate the feature optimization methodology using three real-world matching use cases based on data from economic history research. The use cases (UC) are drawn from distinct data sources and cover different units of analysis (i.e., advertising outlets, companies, and persons), aiming to illustrate the flexibility and applicability of the methodology.

4.1 Data

UC 1. In the first UC, the goal is to match individual news outlets across volumes of the *Zeitungs-Katalog* ([Annoncen-Expedition Rudolf Mosse, 1910–1933](#)). The annually published catalog served as a directory of all advertising outlets (daily newspapers, illustrated papers, trade journals) for which the Annoncen-Expedition Rudolf Mosse agency brokered advertisements. It listed outlets by documenting their headquarter location, operational circulation, political orientation, and advertising prices, providing a detailed map of the German and international press landscape (see Figure A.1 in the Section A for an example snippet.).

UC 2. The second UC involves company data from the *Handbuch der deutschen Aktiengesellschaften* ([1896/97–1918/19](#)). The original sources are digitized and the OCR extracted content is publicly available ([Gram et al., 2022](#), [Universitätsbibliothek Mannheim, 2019](#)). The handbook was published annually since 1896/97 and documented the entire population of German joint-stock companies, both listed and unlisted. A typical handbook record includes the company name, headquarter location, foundation and incorporation dates, equity composition, balance sheet and profit and loss data, as well as the composition of the management and supervisory boards (see Figure A.2, Section A). For larger firms, entries tend to be more extensive, including additional details such as places of stock listing and bond issuance.

UC 3. The third UC concerns (natural) person matching. The data are extracted from the biographical directory *Wer leitet?* ([1940–1941](#)). The directory lists leading figures in German business and public administration, providing information such as names,

addresses, dates and places of birth, and professional positions held. The original sources were digitized, extracted, and enriched with additional metadata variables derived from the original entries (Universitätsbibliothek Mannheim, 2023). Figure A.3 in Section A shows an example page from the source.

Table I provides summary statistics for the three UCs. All cases involve matching entities in data extracted from different volumes of the focal publication series. For each UC, the *Left* dataset (L) contains observations from volume t ; the *Right* dataset (R) contains observations from volume $t + d$ with $d \geq 1$; and the *Matches* dataset (M) records manually verified pairs of indices (ground truth) with matching entities from L and R . The UCs differ markedly in terms of manually verified matching pairs (the ground truth ranges from 175 to 1,484) and available original fields in L and R (between 16 and 1,304). Furthermore, the UCs entail different relationship types between entities in the ground truth. While the person and news outlet UCs contain exactly one observation per real-world entity in both the L and R data sets (1–1 matching), multiple observations of the same company can appear in either data set in the company matching use case (many-many matching). Common to all cases is that the underlying data are non-harmonized because the original sources do not contain standardized entries and their digitized versions potentially include extraction errors. The data for UCs 2 and 3 (persons and companies) were generated using OCR technologies and the data for UC 1 (newspapers) were manually extracted.

4.2 Pre-processing, Feature Optimization, and Classification

The analyses for the three applications are implemented using `neermatch-utilities` and `neermatch` packages (Karapanagiotis and Liebold, 2023).³ For each application, (i) we pre-process the data and construct the candidate feature spaces, (ii) apply the feature optimization methodology of Section 3, and (iii) fit various entity matching models (ANN,

³The packages are available in R and Python and contain functionality commonly used by quantitative economic historians and economists. The `neer-match-utilities` and `neer-match` packages are available on PyPI at <https://pypi.org/project/neer-match-utilities/> and <https://pypi.org/project/neer-match/>, respectively.

Table I: Dataset characteristics and matching configurations

Primary Unit of Analysis	Data Source	Left		Right		#Matches	Field-pairs Used	Rel.	Dirty
		Fields	#Records	Fields	#Records				
Outlet	Zeitungskatalog Mosse	71	1,484	71	1,484	1,484	12	1:1	✓
Company	Handbuch der Deutschen Aktiengesellschaften (HdDAG)	16	692	16	692	692	14	m:m	✓
Person	Wer leitet?	715	175	1,304	175	175	8	1:1	✓

Note: The table summarizes the three data sources used to demonstrate the feature selection procedure. Each matching task involves linking two volumes of the same publication, where *Left* refers to records extracted in volume t and *Right* to records extracted in volume $t + d$, with $d \geq 1$. *Fields* denotes the number of variables in each data set. *#Records* reports the number of observations per data source, and *#Matches* indicates the number of manually verified links between *Left* and *Right*. *Field-pairs Used* refers to the number of distinct field pairs serving as inputs to the feature expansion phase (see Table II for the their combinations with similarity measures). *Rel.* describes the matching relationship, and *Dirty* indicates whether the data contain OCR errors or other imperfections.

Logit, Probit, and Gradient Boosting) using the optimal feature selection. For illustrative purposes, Section B contains the reproducible implementation pipeline (data loading, pre-processing, feature optimization, train-test split, and model training) for the analysis of UC 2 using a subset of features and similarity measures. Below, we describe each component of the pipeline as implemented in the order outlined above.

Pre-processing. We address several issues common to data extracted from historical non-harmonized sources by transforming the data fields to facilitate similarity calculations. The specific transformations vary by use case. For UC 1, we convert textual fields to lowercase and remove standard German stop words using the spaCy `de_core_news_sm` pipeline, a functionality embedded in `neermatch-utilities.Prepare`. For UC 2 and UC 3, we convert textual fields to uppercase. As some fields in the data underlying UC 2 consist of very long entries, we additionally truncate selected long-text fields to 30 characters to reduce noise from verbose free-text entries and lower computational requirements. Across all use cases, missing values are filled with empty alphanumeric values to ensure compatibility with the similarity functions.

Beyond the transformations applied here, the `neermatch-utilities.Prepare` module supports additional preprocessing steps, including the removal of domain-specific stop words (e.g., legal forms such as "Aktiengesellschaft" or "Gesellschaft mit beschränkter Haftung" that may dominate firm name comparisons) and custom spaCy pipelines with

Table II: Field Pairs Across Use Cases

Use Case	Field Pairs
1	newspaper, political, political_harm_cat, newspaper~city, city~newspaper, city, subregion, region, population_log, lat, lng, partial_match
2	company_name, company_info_1, company_info_2, purpose, sboard_text, bs_text, found_year, found_date_modified, register_year, register_date_modified, effect_date_modified, effect_year, city, industry
3	main_info, Vorstand, StVdAR, address, birth_date, raw_text, name, surname

Note: The table lists the field pairs used for entity matching in each use case. Each field pair defines which variables are compared across the two datasets. For most fields, the same variable name appears in both datasets and is compared directly. The symbol $\sim$ denotes a cross-field comparison, where a variable from the left dataset (before $\sim$) is compared against a different variable from the right dataset (after $\sim$). Field pair types are consistent across analysis type within each use case.

user-defined stop word lists.

Feature expansion. The candidate feature spaces are constructed using the field pairs detailed in Table II and similarity measures listed in Table III. We use both measures that are (mathematically) proper normalized similarities with values in $[0, 1]$ and fuzzy string ratios with values potentially outside the interval $[0, 1]$. In the latter case, we rescale the resulting features to $[0, 1]$ for numerical stability and consistency in interpretations. For fields with multiple missing observations, we introduce similarity measures that reward record pairs based on data completeness: The `neermatch-utilities.notmissing` similarity measure returns 1 if both values in a record pair are present (i.e., neither `None`, an empty string, nor `NaN`), and 0 otherwise. Similarly, `neermatch-utilities.notzero` returns 1 if both numeric values are present and non-zero, and 0 otherwise. These measures prove particularly valuable when working with sparse historical datasets where missingness patterns carry information in themselves.⁴

Feature optimization. Across all UCs and analyses, we use an identical feature

⁴For textual fields representing names, `neermatch-utilities.Commonness` also implements a frequency-based similarity measure that rewards agreement on rare values (e.g., uncommon surnames) while down-weighting agreement on frequent ones (see also Stringham, 2022). We do not use this measure in the analyses below to keep the comparison consistent and focused on standard similarity measures.

Table III: Similarity Measures per Field-pair Type across Use Cases.

Similarity Measures	Use Case 1		Use Case 2		Use Case 3	
	(1)	(2)	(1)	(2)	(1)	(2)
<i>Alphanumeric Field Pairs</i>						
levenshtein, jaro_winkler, prefix, postfix, token_sort_ratio, token_set_ratio, partial_token_set_ratio, partial_token_sort_ratio, lcsseq, jaro	✓	✓	✓	✓	✓	✓
notmissing	✓	✓	✓	✓	✓	
basic_ratio, damerau_levenshtein, hamming, indel, lcsseq, osa, partial_ratio, partial_token_ratio, token_ratio	✓		✓		✓	
<i>Numeric Fields Pairs</i>						
notzero	✓	✓	✓	✓		
euclidean	✓	✓				
gaussian	✓		✓			
<i>All Fields Pairs</i>						
discrete	✓	✓	✓	✓	✓	✓

Note: The table illustrates the set of used similarity concepts across the use cases. It distinguishes between two analysis types. Type (1) refers to the regression-based analysis using a logit classifier, whose results are reported in Table V. Type (2) refers to Figure II, which illustrates heterogeneous performance gains from feature optimization across classifier types. For (1), the listed similarities constitute the candidate pool available for feature optimization; for (2), they reflect the full used similarity map specification. Numeric similarities (**notzero**, **euclidean**, **gaussian**) were only applied where numeric field pairs were available or selected.

selection configuration. In the first stage, we filter out nearly collinear features using a correlation clustering threshold $\tau = 0.95$. In the second stage, we estimate elastic net-regularized logistic models via stratified 2-fold cross-validation. The regularization strength C is searched over a grid of 20 logarithmically spaced values in $[10^{-2}, 10^3]$, and the ℓ_1 ratio λ is searched over $\{0.8, 0.9, 1.0\}$. The parameter combination that maximizes the average precision score is selected, and the final model is refit on the full training set. Features with absolute coefficients $|\beta_j|$ below 0.01 are discarded.

Classification. The labeled match pairs are split into training, validation, and test sets. The training share is 30%, 50%, and 60% for UC 1, UC 2, and UC 3, respectively; the test share is 60%, 40%, and 30%. A validation share of 10% is reserved across all

use cases and classifiers to ensure that all models are trained on exactly the same data; however, only the gradient boosting classifier uses the validation set, for threshold tuning via the Matthews correlation coefficient (MCC). Table IV summarizes the training hyperparameters for all classifiers.

The logit, probit, and gradient boosting classifiers are trained on the cross-product of observations of the left and right (training) datasets, i.e., $n = n^L \cdot n^R$, including all mismatches. The ANN models are trained using pre-training and fine-tuning stages to mitigate class imbalance effects. The pre-training stage minimizes a binary cross-entropy loss for 300 epochs on a subset of records where mismatches are severely sub-sampled, including only a share of 0.05 of all non matches. Subsequently, the models are fine-tuned for 50 epochs on the complete training set using focal loss minimization (Lin et al., 2017) with a focusing parameter $\gamma = 0$. This reduces to an α -weighted cross-entropy, where the positive-class weight α is computed as $\alpha = s \cdot N_- / (s \cdot N_- + N_+)$, where s is the non-match share, N_- is the number of non-match pairs, and N_+ is the number of match pairs, and is capped at $\alpha_{\max} = 0.85$.⁵

4.3 Results and Evaluation

4.3.1 Feature Optimization Impact on Logit Matching Performance

We investigate how feature expansion and reduction affect matching performance across the matching cases. Given the large number of models to be estimated, we rely on a Logit classifier throughout this subsection for computational efficiency. We start by drawing random samples that vary both in the number and composition of field pairs (ranging from 1 to 12) and similarity measures (ranging from 1 to 24) from the news outlet data (UC 1). We refer to each random draw of field pairs, similarity measures, and train-test

⁵All trained models can be used to construct longitudinal panels from repeated cross-sections via the package’s `GenerateID` class. It compares records across consecutive time periods, filters matches by a configurable prediction threshold, and chains matches across periods to assign consistent panel identifiers. To reduce computational costs in large datasets, researchers can partition the data into subgroups that act as blocking variables (e.g., by geographic region). The class supports four relationship types (`1:1`, `1:m`, `m:1`, `m:m`), enabling users to control whether records are matched uniquely or allowed to match to multiple counterparts.

Table IV: Training hyper-parameters by classifier.

Parameter	Logit	Probit	GB	ANN
<i>Data splitting</i>				
Training share (UC 1 / UC 2 / UC 3)	0.30 / 0.50 / 0.60	0.30 / 0.50 / 0.60	0.30 / 0.50 / 0.60	0.30 / 0.50 / 0.60
Validation share	0.10	0.10	0.10	0.10
Test share (UC 1 / UC 2 / UC 3)	0.60 / 0.40 / 0.30	0.60 / 0.40 / 0.30	0.60 / 0.40 / 0.30	0.60 / 0.40 / 0.30
<i>Training</i>				
Non-match share (fitting)	1.0	1.0	1.0	—
Threshold tuning metric	—	—	MCC	—
<i>ANN architecture</i>				
Feature network width scale / depth	—	—	—	2 / 4
Record network width scale / depth	—	—	—	2 / 4
<i>ANN Stage 1 (pre-training)</i>				
Loss function	—	—	—	Binary cross-entropy
Epochs	—	—	—	300
Non-match share	—	—	—	0.05
<i>ANN Stage 2 (fine-tuning)</i>				
Loss function	—	—	—	Focal ($\gamma=0.0$, $\alpha_{\max}=0.85$)
Epochs	—	—	—	50
Non-match share	—	—	—	1.0

splits as a *data fold*. We generate 1,000 such folds in total. For each fold, we fit two logit models: one applying feature selection and one without. While the train-test splits vary randomly across folds by design, they are held constant for the two models trained to ensure comparability and isolate the effects of feature expansion and selection. Each model is trained on 30 percent of the ground truth and evaluated on 60 percent of the manually labeled data. Thus, each model yields its own training and test performance statistics. To examine the effect of the two components of feature optimization on test performance, we calculate and use the F1 score as the dependent variable in regressions progressively targeting the specification

$$\begin{aligned}
F1_h = & \beta_0 + \beta_1 \text{FieldPairs}_h + \beta_2 \text{DistConcepts}_h + \beta_3 \text{FeatSel}_h \\
& + \beta_4 (\text{FieldPairs}_h \times \text{FeatSel}_h) + \beta_5 (\text{DistConcepts}_h \times \text{FeatSel}_h) + \varepsilon_h,
\end{aligned} \tag{1}$$

where FieldPairs_h denotes the number of field pairs, DistConcepts_h the number of similarity measures used per field pair (i.e., a categorical measure of feature expansion), and FeatSel_h a binary indicator for whether feature selection was applied in a given model. Equation 1 represents the most comprehensive specification. We estimate a sequence of models, beginning with a baseline that includes only FieldPairs_h and DistConcepts_h , then progressively adding the feature selection indicator and its interactions. Panel 1 of

Table V reports the results across all five specifications.

Importantly, a first finding is that all specifications suggest that increasing both the number of field pairs and the number of similarity measures has a positive effect on matching performance. An additional field pair is estimated to increase the F1 score by between 0.038 points (Column 5) and 0.042 points (Column 1). Relative to the sample mean F1 score of 0.41, this represents an average increase of approximately 9.3 to 10.2 percent. While likewise consistently positive and statistically significant, the estimated performance increase from extending the model by one similarity measure is somewhat smaller, ranging between 1.9 and 3.0 percent relative to the sample mean.

With regards to feature selection, the coefficient in Column 2 suggests that following the methodology of Section 3 and actively reducing the number of features while holding the number of field pairs and similarity measures constant increases the test data F1 score by 0.039 points on average (an increase of approximately 9.4 percent relative to the sample mean). However, the estimated coefficients from Columns 3 to 5 reveal that this average effect masks substantial heterogeneity: the impact of feature selection is highly dependent on the number of field pairs and similarity measures in the model. While performing feature selection in simple models with few field pairs and similarity measures reduces the F1 score substantially, performance benefits markedly from feature selection as the number of features increases. The additive nature of the interaction effects allows us to illustrate this by means of examples. Assuming a model with three field pairs and two similarity measures, feature selection reduces the F1 score on average by 0.068 points ($= -0.1156 + 3 \times 0.0072 + 2 \times 0.0128$). Conversely, in a model with eight field pairs and 15 similarity measures, feature selection improves the already considerably higher starting F1 score by an additional 0.134 points ($= -0.1156 + 8 \times 0.0072 + 15 \times 0.0128$). This increase represents a 33 percent improvement in performance relative to the sample mean.

The analyses of UCs 2 and 3 largely yield corroborating findings (Panels 2 and 3 of Table V). Across both UCs, the number of field pairs and similarity measures retains a positive and statistically significant effect on matching performance, with the exception

Table V: Matching Performance with and without Feature Optimization.

	(1)	(2)	(3)	(4)	(5)
Panel 1: Outlets					
FieldPairs	0.0424*** (0.0014)	0.0428*** (0.0014)	0.0362*** (0.0018)	0.0406*** (0.0014)	0.0381*** (0.0017)
DistConcepts	0.0116*** (0.0008)	0.0125*** (0.0008)	0.0120*** (0.0008)	0.0079*** (0.0009)	0.0083*** (0.0009)
FeatSel		0.0386*** (0.0087)	-0.0562*** (0.0172)	-0.0922*** (0.0159)	-0.1156*** (0.0186)
FieldPairs $\times$ FeatSel			0.0173*** (0.0027)		0.0072** (0.0030)
DistConcepts $\times$ FeatSel				0.0146*** (0.0015)	0.0128*** (0.0017)
Adj. R-squared	0.515	0.520	0.532	0.546	0.548
Dep. Var Mean	0.410	0.410	0.410	0.410	0.410
Dep. Var SD	0.238	0.238	0.238	0.238	0.238
N	1626	1626	1626	1626	1626
Panel 2: Companies					
FieldPairs	0.0517*** (0.0019)	0.0515*** (0.0019)	0.0534*** (0.0026)	0.0500*** (0.0019)	0.0533*** (0.0026)
DistConcepts	0.0132*** (0.0014)	0.0144*** (0.0014)	0.0146*** (0.0015)	0.0100*** (0.0018)	0.0101*** (0.0018)
FeatSel		0.0531*** (0.0155)	0.0804*** (0.0301)	-0.0549* (0.0309)	-0.0156 (0.0371)
FieldPairs $\times$ FeatSel			-0.0040 (0.0037)		-0.0072* (0.0038)
DistConcepts $\times$ FeatSel				0.0121*** (0.0030)	0.0133*** (0.0031)
Adj. R-squared	0.500	0.506	0.506	0.514	0.515
Dep. Var Mean	0.723	0.723	0.723	0.723	0.723
Dep. Var SD	0.328	0.328	0.328	0.328	0.328
N	934	934	934	934	934
Panel 3: Persons					
FieldPairs	0.0607*** (0.0031)	0.0706*** (0.0031)	0.0637*** (0.0037)	0.0702*** (0.0031)	0.0638*** (0.0037)
DistConcepts	-0.0002 (0.0012)	0.0067*** (0.0013)	0.0066*** (0.0013)	0.0049*** (0.0014)	0.0049*** (0.0014)
FeatSel		0.1497*** (0.0147)	0.0661** (0.0283)	0.0819*** (0.0261)	0.0105 (0.0344)
FieldPairs $\times$ FeatSel			0.0236*** (0.0069)		0.0219*** (0.0069)
DistConcepts $\times$ FeatSel				0.0113*** (0.0036)	0.0103*** (0.0036)
Adj. R-squared	0.175	0.216	0.220	0.219	0.223
Dep. Var Mean	0.849	0.849	0.849	0.849	0.849
Dep. Var SD	0.293	0.293	0.293	0.293	0.293
N	1970	1970	1970	1970	1970

Note: OLS regression results with the F1 score as the dependent variable. Sample sizes are report at the bottom of each panel. For each data point, two Logit classifiers are trained: one with and one without feature selection. Train-test splits are held constant within each fold to ensure comparability. Each model is trained on 30 percent and evaluated on 60 percent of the manually labeled data. Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

of DistConcepts in the baseline specification of UC 3, where the coefficient is near zero and insignificant before the inclusion of the feature selection indicator. The interaction between similarity measures and feature selection is consistently positive and significant across all three cases, confirming that the benefits of feature selection grow with the dimensionality of the feature space. The interaction between field pairs and feature selection, however, is less stable: while positive and significant in UC 1 and UC 3, it is slightly negative and marginally significant in UC 2, suggesting that in the company matching case, adding field pairs does not amplify (or may even slightly attenuate) the gains from feature selection. Notably, the unconditional feature selection coefficient in Column 2 is considerably larger in UC 3 (0.150) than in UC 1 (0.039) and UC 2 (0.053), indicating that feature selection is particularly beneficial for person matching even in moderately sized models. As in UC 1, the full specification (Column 5) reveals that this average effect masks heterogeneity driven by the interaction terms.

Overall, the results reinforce the conclusion that feature selection is not universally beneficial but becomes increasingly valuable as model complexity grows, with the similarity measure dimension being the more robust driver of this pattern across use cases.

Intuitively, in models with weak feature expansion, that is, with few similarity measures per field pair, feature selection risks removing relevant information, thereby reducing performance. Conversely, in models with a large feature space, feature selection helps to eliminate noise and redundancy, thereby increasing model performance substantially. In total, the best-performing models are those that combine strong feature expansion with subsequent feature selection.

4.3.2 Performance Heterogeneity across Classification Models

The previous section examined under which conditions and by how much feature optimization affects matching performance, focusing exclusively on Logit models. We now extend the analysis by comparing performance effects of feature selection across four classifier types: Logit, Probit, Gradient Boosting, and ANNs across UCs. Each UC analysis comprises 50 data folds, with three model variants trained per fold and classifier, yielding

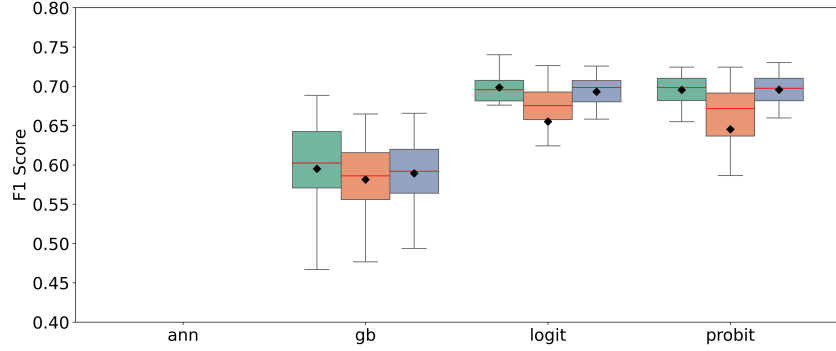
600 estimated models ($4 \times 50 \times 3$) per matching UC. All models within a given case operate on the same set of feature pairs (12 for newspapers, 14 for companies, and 8 for persons; see Table II) but differ in the number of similarity measures employed. *Reference models* rely on a single Jaro-Winkler-based similarity measure. *Expanded models* utilize a larger set of similarity measures (14 for UC 1, 13 for UC 2, and 11 for UC 3; see Table III) without feature selection. *Reduced models* are identical to the expanded models except that feature selection is applied.⁶ To ensure comparability, we hold the train-test split constant across all model variants and classifier types within a given data fold. That is, the reference, expanded, and reduced variants of every classifier are trained and evaluated on exactly the same data partition. Across folds, the split varies, but within each fold it remains identical for all models.

Figure IIb illustrates the distribution of F1 scores across classifiers and model variants for the company matching UC. Box plots are grouped by classifier (ANNs, Gradient Boosting, Logit, and Probit, from left to right), with colors indicating the model variant: green for reference, orange for extensive, and purple for reduced. Within each box plot, the red horizontal line marks the median and the black square marks the mean F1 score. The colored boxes span the interquartile range (Q1–Q3), and the whiskers extend to 1.5 times the IQR.

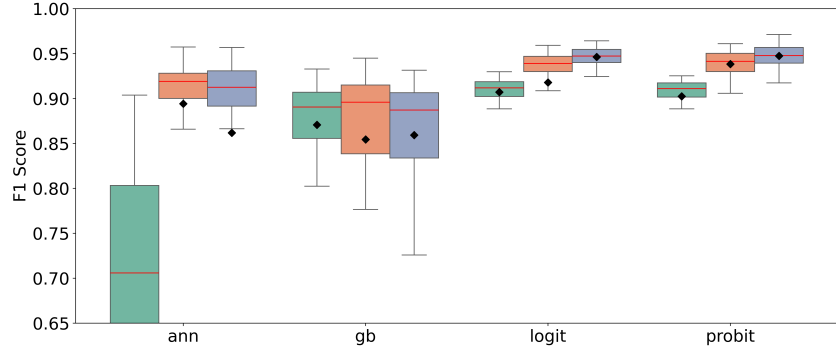
Figure IIb reveals four findings. First, linear Probit and Logit models outperform both Gradient Boosting and ANNs on average. This result is broadly in line with Feigenbaum (2016), who finds that Probit and Logit outperform Random Forest and SVM in entity matching for historical census. Our findings extend this result by showing that linear classifiers are also capable of outperforming non-linear methods in the context of historical firm-level entity matching. Second, ANN models trained on the Jaro–Winkler similarity alone, without feature selection, exhibit notably inconsistent performance with high variance in F1 scores. Although the extended and reduced models outperform the reference model in this use case, inconsistent ANN performance across other use cases (see below) and substantially longer estimation times raise concerns about the practical

⁶Note that even for the reduced and expanded models, we employ only a subset of the available similarity measures utilized above, due to computational constraints.

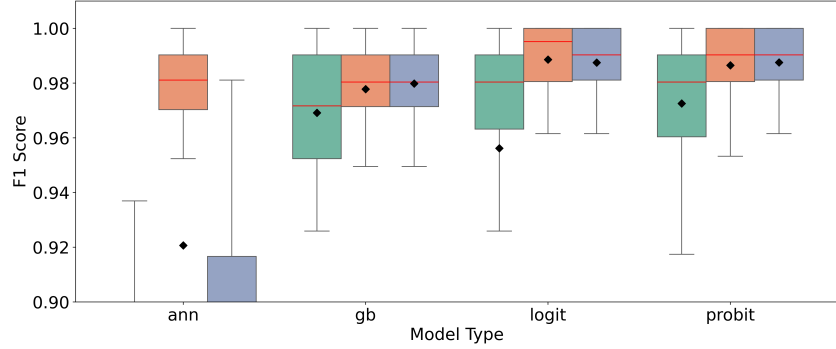
Figure II: Overall caption for all subfigures.



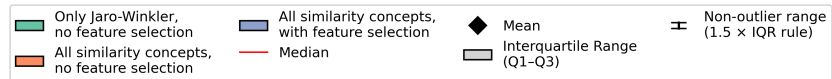
(a) UC 1: Newspaper



(b) UC 2: Companies



(c) UC 3: Persons



Note: Distribution of F1 scores on test data across up to four classifiers (ANNs, Gradient Boosting, Logit, and Probit) and three model variants for each use case. Each classifier is evaluated over 50 data folds with varying train-test splits, yielding 150 estimations per classifier. Green boxes denote reference models using only Jaro-Winkler similarity, orange boxes denote expanded models using 11 similarity concepts without feature selection, and purple boxes denote reduced models applying feature selection to the expanded feature set. All model variants within a given data fold share the same train-test split. Red horizontal lines indicate medians, black diamonds indicate means, boxes span the interquartile range (Q1–Q3), and whiskers extend to 1.5 times the IQR. ANNs were not evaluated for the newspaper use case due to insufficient training data.

applicability of ANNs in economic history. In contrast Logit and Probit models tend to be much more consistent, yielding narrower interquartile ranges. Third, as measured by median F1 scores, incorporating multiple similarity measures consistently outperforms relying solely on Jaro-Winkler-based metrics across all classifiers.

A fourth central finding is that feature selection proves beneficial primarily for linear models: for Logit, the mean F1 score increases from 0.918 to 0.946, while for Probit, the highest realized F1 score rises from 0.961 to 0.971. Conversely, no comparable improvements emerge for ANNs or Gradient Boosting. These observed patterns suggest that differences in the complexity of the classifier do not fully account for variation in matching performance. Instead, performance appears closely tied to how features are constructed and combined.

Since non-linear models are inherently less sensitive to multicollinearity than linear ones (e.g., [Chowdhury et al., 2022](#)), the performance gains from feature selection in linear models likely stem from reducing highly correlated variables rather than removing irrelevant ones. In other words, the first step of the feature selection procedure seems to have a greater impact on performance than the second. This conclusion rests on the reasoning that, if the removal of irrelevant variables were the driving factor, we would expect significant performance increases across all model types.

Moreover, the previous subsection showed that the efficiency of feature selection grows with the number of feature pair and similarity measure combinations. Given that computational limitations restricted us to only a subset of the available similarity measures for the company matching use case, compared to 24 in the first task, the observed performance gains from feature selection likely understate the maximum potential for at least regression-based models.

Results from the person matching applications largely support these conclusions. As with company matching, median and mean F1 scores associated with UC 3 are higher when incorporating multiple similarity measures compared to relying solely on the Jaro-Winkler distance across all classifiers (see [Figure IIc](#)). Additionally, only the expanded and reduced linear models exhibit interquartile ranges that includes a perfect F1 score

of 1 on the test data. Assessing whether feature selection further improves performance is difficult, however, as the expanded models already frequently achieve perfect scores, reflecting the relative simplicity of this matching use case. What is apparent, however, is that feature selection drastically reduces the performance of ANN-based models in this matching use case. Beyond the concerns about straightforward ANN use when relying on one similarity measures, this finding also calls into question the general suitability of feature selection for this type of model.

For the newspaper matching case (UC 1), the results differ somewhat. Models relying solely on the Jaro-Winkler distance perform comparably to the reduced models, and both outperform the expanded models across most classifier types (Figure IIa).⁷ However, as discussed above, efficiency gains from feature selection tend to increase with the number of similarity measures employed. The fact that reduced models do not outperform the reference models here may therefore stem from the limited set of similarity measures available for selection in this exercise.

5 Conclusions

In this article, we presented a reproducible and transparent methodology for optimizing feature selection in entity matching, addressing the needs of scholars working with historical datasets. We found that the methodology improved performance across three distinct matching applications in two key ways. First, employing a wide range of similarity measures (up to 24) improves matching performance across all tested classifiers, including Logit, Probit, Gradient Boosting, and ANNs. Second, for linear classifiers, a subsequent feature selection step further improves performance by removing highly correlated and uninformative features. By providing a tractable implementation through **neermatch** and **neermatch-utilities**, we equip researchers with tools to identify optimal feature sets tailored to their specific matching tasks.

These findings suggest that improvements in entity matching for historical source materials are less likely to arise from increasingly advanced algorithms alone, and more

⁷Due to computational limitations and the size of the available labeled training data, we did not train ANN-based classifiers.

from methodologies that make the construction, evaluation, and selection of features explicit, transparent, and reproducible. Future research would benefit from developing systematic approaches to diagnosing and mitigating the biases introduced by the particularities of historical source materials, thereby improving the reliability of matched historical datasets.

Data availability

In accordance with the journal’s data availability policy, the authors are willing to deposit the code and data required to reproduce all results reported in this article in the OPEN ICPSR repository upon publication. The `neer-match` and `neer-match-utilities` packages are openly available on PyPI at <https://pypi.org/project/neer-match/> and <https://pypi.org/project/neer-match-utilities/>, respectively.

Declaration of generative AI use

During the preparation of this work, the author(s) used LLMs in order to improve language and readability. After using this tool, the author(s) reviewed and edited the content as needed and take full responsibility for the content of the publication.

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